

ANTIDERIVATIVE

If $F'(z) = f(z)$ then $\int_{z_1}^{z_2} f(z) dz = F(z_2) - F(z_1)$

$\int f(z) dz = F(z)$

Example: $\int_0^{2+i} z^3 dz = \left[\frac{z^4}{4} \right]_0^{2+i}$

$= \frac{1}{4} [(2+i)^4 - 0^4]$

$= \frac{1}{4} (16 + 32i - 24 - 8i + 1)$

$= \frac{1}{4} (24i - 7)$

$(x+y)^4$
 $= x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$

Considered
 $i^2 = -1$, $i^3 = -i$
 $i^4 = 1$

Let a function f of a complex variable is continuous in a domain D . Then there exists a function F such that $F'(z) = f(z)$, $\forall z \in D$ is called an antiderivative of f . i.e. $\int f(z) dz = F(z)$.

Complex Valued function on Real Domain:
 $\left\{ \begin{array}{l} f: D \rightarrow \mathbb{C} \\ \text{domain} \end{array} \right.$ while $D \subset \mathbb{R}$
 $\hookrightarrow$ complex variable

$f(t) = u(t) + iv(t)$, $t \in [a, b]$

Definite Integral

$\int_a^b f(t) dt = \int_a^b u(t) dt + i \int_a^b v(t) dt$

Example 1. $f(t) = t^2 + 1 + it^3$; $0 \leq t \leq 1$

$$\begin{aligned} & \int_0^1 (t^2 + 1) dt + i \int_0^1 t^3 dt \\ &= \left[\frac{t^3}{3} + t \right]_0^1 + i \left[\frac{t^4}{4} \right]_0^1 \\ &= \frac{1}{3} + 1 + i \frac{1}{4} = \frac{4}{3} + \frac{1}{4}i \end{aligned}$$

$$z = \begin{cases} x+iy & \leftrightarrow \text{standard form} \\ r(\cos\theta + i\sin\theta) \\ re^{i\theta} \end{cases}$$

Alternatively

2. $f(t) = e^{i2t}$; $0 \leq t \leq \pi/6$

$$\begin{aligned} \int_0^{\pi/6} e^{i2t} dt &= \int_0^{\pi/6} \cos 2t dt + i \int_0^{\pi/6} \sin 2t dt \\ &= \left[\frac{\sin 2t}{2} \right]_0^{\pi/6} + i \left[-\frac{\cos 2t}{2} \right]_0^{\pi/6} \\ &= \frac{1}{2} \left[\sin 2\left(\frac{\pi}{6}\right) - \sin 2(0) \right] + i \left(-\frac{1}{2} \right) \left[\cos 2\left(\frac{\pi}{6}\right) - \cos 2(0) \right] \end{aligned}$$

$$\begin{aligned} \int e^{i2t} dt &= \frac{e^{i2t}}{2i} \cdot \left(\frac{-i}{-i} \right) \\ &= \dots e^{i2t} \end{aligned}$$

$= x+iy$
present in
standard form

$$= \frac{1}{2} \sin \frac{\pi}{3} - \frac{i}{2} \cos \frac{\pi}{3} + \frac{i}{2} \cos(0)$$

$$= \frac{1}{2} \cdot \frac{\sqrt{3}}{2} - \frac{i}{2} \cdot \frac{1}{2} + \frac{i}{2} \cdot 1$$

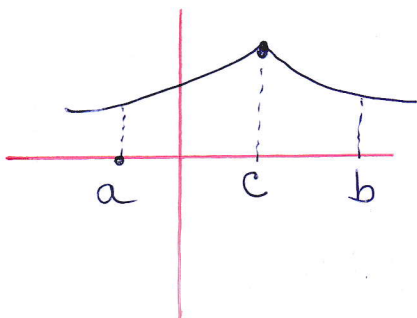
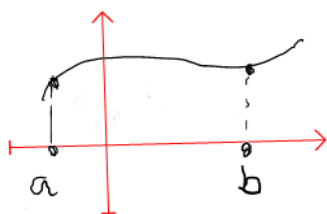
$$= \frac{\sqrt{3}}{4} - i \frac{1}{4} + i \frac{1}{2} = \frac{\sqrt{3}}{4} + i \frac{1}{4}$$

Properties:

Let $f(t) = u(t) + iv(t)$, $g(t) = p(t) + iq(t)$
 $t \in [a, b]$;

$$(i) \int_a^b [f(t) \pm g(t)] dt = \int_a^b f(t) dt \pm \int_a^b g(t) dt$$

$$(ii) \int_a^b f(t) dt = \int_a^c f(t) dt + \int_c^b f(t) dt, \quad c \in [a, b]$$



when there is
no direct
path between
a and b.

$$(iii) \int_a^b f(t) dt = - \int_b^a f(t) dt$$

$$(iv) \int_a^b f(t) g(t) dt = \int_a^b (u + iv)(p + iq) dt$$

$$= \int_a^b (up + ivp + iuq + i^2 vq) dt$$

$$= \int_a^b (up - vq) dt + i \int_a^b (vp + uq) dt$$

$$(v) \operatorname{Re} \int_a^b f(t) dt = \int_a^b \operatorname{Re}(f(t)) dt$$

$$(vi) \int_a^b z_0 f(t) dt = \int_a^b (x_0 + iy_0)(u + iv) dt$$

Complex
Constant
 $z_0 = x_0 + iy_0$

$$= \int_a^b (x_0 u + i y_0 u + i x_0 v + i^2 y_0 v) dt$$

$$= z_0 \int_a^b [u(t) dt + i v(t) dt]$$

$$= z_0 \left[\int_a^b u(t) dt + i \int_a^b v(t) dt \right]$$

$$= (x_0 + iy_0) \left[\int_a^b u(t) dt + i \int_a^b v(t) dt \right]$$

(vii) Absolute Value of integral:

$$f(t) = u + iv$$

$$\therefore |f(t)| = |u + iv| = \sqrt{u^2 + v^2}$$

$$\left| \int_a^b f(t) dt \right| \leq \int_a^b |f(t)| dt$$

Improper Integrals

$$\left| \int_a^b (u+iv) dt \right| \leq \int_a^b \sqrt{u^2+v^2} dt$$

while there is no direct path or there exist asymptote

$$f: \underset{\text{domain}}{D} \longrightarrow \underset{\text{Complex Variable}}{C}, \quad D \subset \mathbb{R}$$

$$f(t) = u(t) + iv(t), \quad t \in \mathbb{R}$$

$$\int f(t) dt = \int u(t) dt + i \int v(t) dt$$

Provided that the right side integral exist

All properties of real integrals hold true

Indefinite to Definite Integral:

$$\text{Let } U'(t) = u(t), \quad V'(t) = v(t)$$

$$f(t) = u(t) + iv(t)$$

$$\text{Then } \int u(t) dt = U(t), \quad \int v(t) dt = V(t)$$

$$\therefore \int f(t) dt = U(t) + iV(t)$$

$$\text{Then } \int_a^b f(t) dt = \left[U(t) + iV(t) \right]_a^b$$

$$= U(b) - U(a) + i(V(b) - V(a))$$

$$= U(b) + iV(b) - (U(a) + iV(a))$$

$$= F(b) - F(a)$$

$$\int f(t) dt = F(t)$$

**See Exapmle 1 & 2
on page 11 & 12**

Paths: arcs & curves

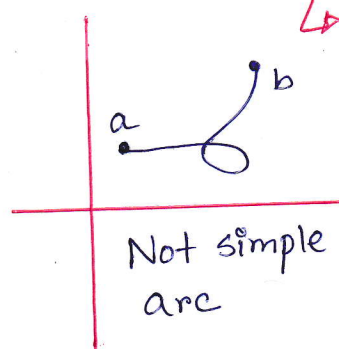
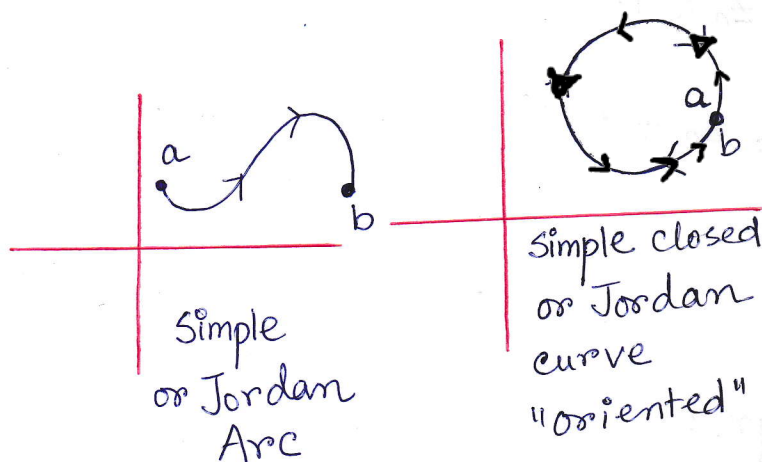
An arc C in complex plane

$$z(x, y), \quad x = x(t), \quad y = y(t), \quad a \leq t \leq b$$

- t is a free parameter
- t represents characteristic of an equation.

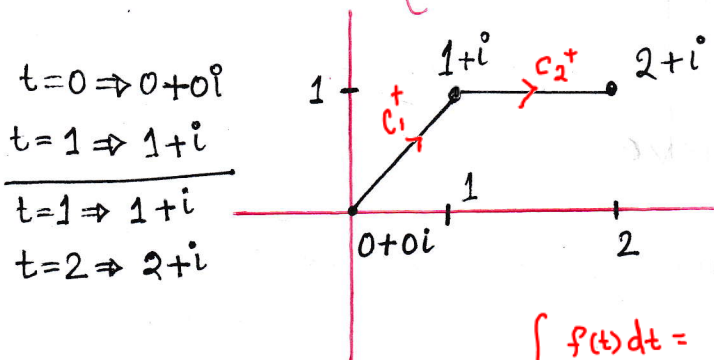
Eqn of arc: $z = z(t) = x(t) + iy(t), \quad a \leq t \leq b$ ex

$$\begin{aligned} x^2 + y^2 &= r^2 \\ r^2 \cos^2 \theta + r^2 \sin^2 \theta &= r^2 \\ \cos^2 \theta + \sin^2 \theta &= 1 \\ \hookrightarrow \text{eqn of circle.} \end{aligned}$$



Examples

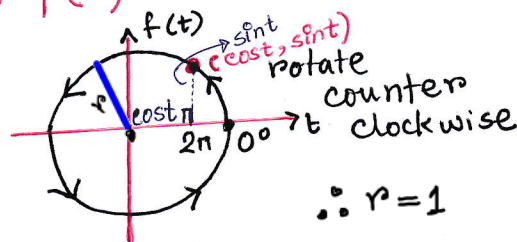
$$1. f(t) = \begin{cases} t + it, & 0 \leq t \leq 1 \\ t + i, & 1 \leq t \leq 2 \end{cases}$$



$$\int_C f(t) dt = \int_{C_1} f(t) dt + \int_{C_2} f(t) dt = \text{Ans}$$

$C \in (0, 2)$ $C_1 \in (0, 1) + C_2 \in (1, 2)$

$$2. f(t) = e^{it}, \quad 0 \leq t \leq 2\pi$$



$$e^{it} = \cos t + i \sin t$$

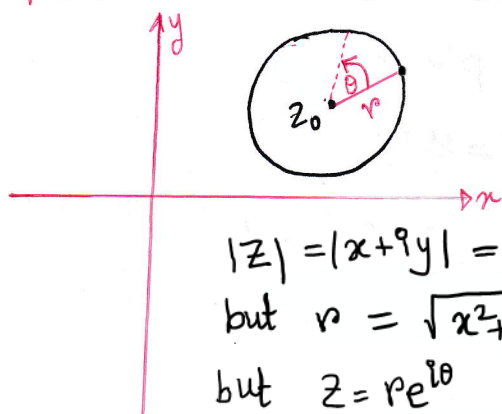
Standard form $z = z_0 + re^{i\theta}$ but $z_0 = (0, 0)$

Circle Center: (0, 0) Radius = r

$\therefore z = 0 + re^{i\theta}$

$z = re^{i\theta}$

3. $f(t) = z_0 + re^{it}$, $0 \leq t \leq 2\pi$



$$Z = \begin{cases} x+iy \\ r(\cos\theta + i\sin\theta) \\ re^{i\theta} \end{cases}$$

$$|Z| = |x+iy| = \sqrt{x^2+y^2}$$

$$\text{but } r = \sqrt{x^2+y^2} \quad \therefore r = |Z|$$

$$\text{but } Z = re^{i\theta} \quad \text{exponential form}$$

$$Z = z_0 + re^{i\theta} \quad \text{eqn of circle}$$

$z_0 = \text{Center of the circle}$

$$Z = re^{i\theta} \Rightarrow z_0 = 0$$

$$Z = z_0 + re^{i\theta} \Rightarrow z_0 \neq 0$$

Curves

Consider parametric eqn:

$$Z = Z(t) = x(t) + iy(t); \quad t_1 \leq t \leq t_2$$

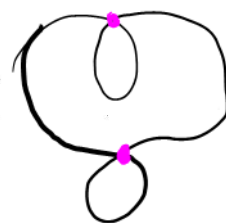
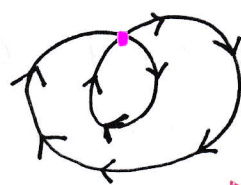
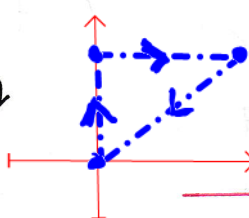
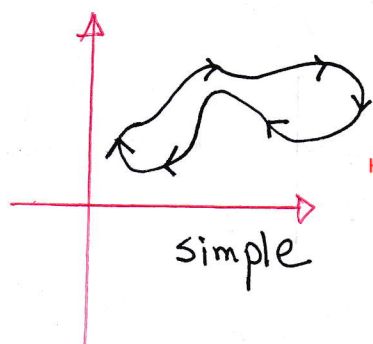
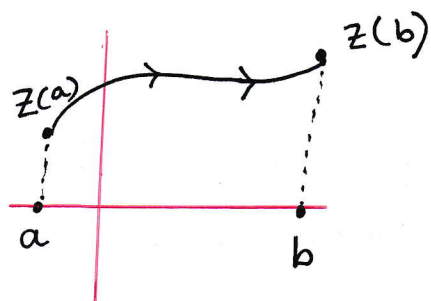
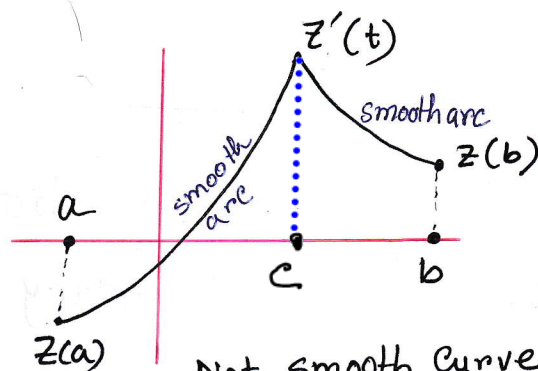


Figure: Closed Curve

A closed curve which does not intersect itself anywhere is called a simple closed curve.



smooth curve



Not smooth curve

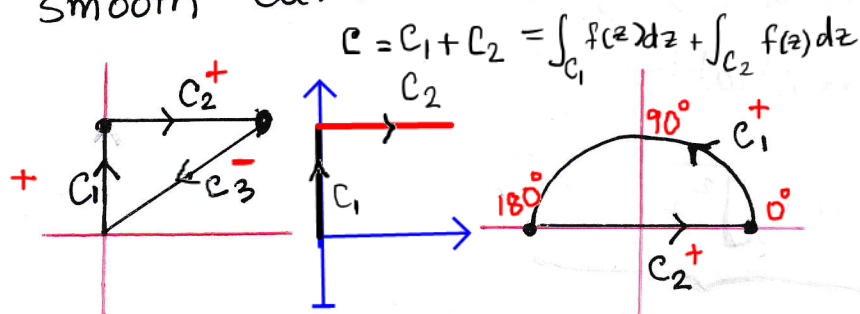
$z(t)$ or $(x(t), y(t))$
has continuous derivatives
in $a \leq t \leq b$.

It is a piecewise
or sectionally smooth
curve.

Smooth arc:
 $\rightarrow z(t)$ is smooth if $z'(t)$ is continuous and $z'(t) \neq 0$,
 $\forall t \in [a, b]$

Contour or Path

When C is constructed by joining finitely many
smooth curves end to end is called a contour.



$$C = C_1 + C_2 + C_3$$

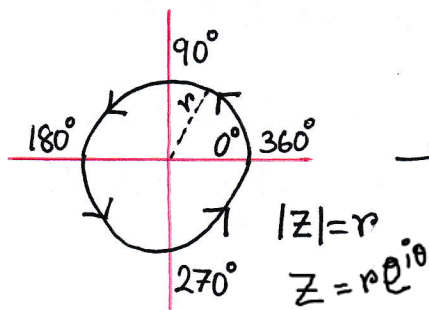
$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz + \left[- \int_{C_3} f(z) dz \right]$$

$$C = C_1 + C_2 + C_3 + \dots + C_n$$

C could be a combination of n many smooth curves

Contour Integral

$\oint_C f(z) dz \rightarrow$ integration of $f(z)$ around the boundary C in a 'tve' sense.
(counter clockwise)



$\rightarrow$ the integral around C is called contour integral.

See Example on Page 14 & Exercise 4 Problem on page 17, 18, 19

Length of Arc

Let $Z = Z(t) = x(t) + iy(t) ; a \leq t \leq b$

$\hookrightarrow$ arc on curve

Suppose $x'(t)$ and $y'(t)$ exist and are continuous throughout the interval $a \leq t \leq b$

$\therefore Z'(t) = x'(t) + iy'(t) ; a \leq t \leq b$

$Z(t)$ is differentiable over $[a, b]$

$$|Z'(t)| = \sqrt{[x'(t)]^2 + [y'(t)]^2}$$

$$\begin{aligned} \because Z &= x + iy \\ \Rightarrow |Z| &= \sqrt{x^2 + y^2} \end{aligned}$$

It is integrable on $[a, b]$

$$\therefore \text{Length of arc } L = \int_a^b |Z'(t)| dt$$

$$= \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2} dt$$

Recall MAT120
Length of parametric curve

$$L = \int \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

See page 13

LINE Integral

$\int_{C_1} f(z) dz = \int_{C_2} f(z) dz$

$\int_{C_1+C_2} f(z) dz = \int_{C_3} f(z) dz = \int_{C_4} f(z) dz$

$C_1 \leq C_2$
 C_1 & C_2
 are independent
 path

Independent
 PATH

$C_1 + C_2 \geq C_3$
 $C_1 + C_2, C_3, C_4$ are all
 independent of path

Cauchy's Theorem or Cauchy Goursat Theorem

If $f(z)$ is analytic everywhere within a simply connected region then $\oint_C f(z) dz = 0$ for every

simple curve 'C' lying in the region R ,

since $\oint_C \dots dz = \int [C_1 + C_2 + C_3 + \dots + C_n] = 0$

$\int_{C_1} f(z) dz + [-\int_{C_2} f(z) dz] = 0$

Completing full turn
 & coming back to
 original point.

Path Independence:

If $f(z)$ has an antiderivative in an open region A , then the path integral $\oint_C f(z) dz$ is path independent for all paths in A .

The following TWO things are equivalent:

- (i) The integral $\oint_C f(z) dz$ is path independent
- (ii) The integral $\oint_C f(z) dz$ around any closed path is 0.

Proof

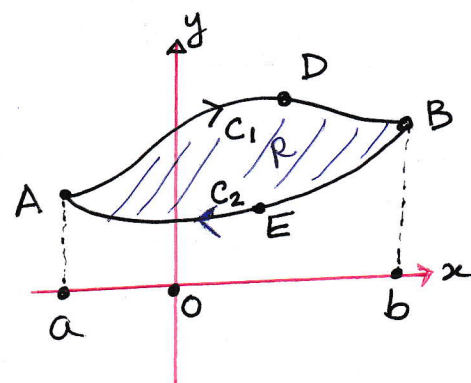
By Cauchy's Theorem:

$$\int_{ADBEA} f(z) dz = 0$$

$$\Rightarrow \int_{ADB} f(z) dz + \int_{BEA} f(z) dz = 0$$

$$\Rightarrow \int_{ADB} f(z) dz = - \int_{BEA} f(z) dz = \int_{AEB} f(z) dz$$

$$\Rightarrow \int_{C_1} f(z) dz = \int_{C_2} f(z) dz = \int_a^b f(z) dz$$



$C_1 \rightarrow$ heading to +ve direction

$C_2 \rightarrow$ heading to -ve direction

Hence the integral is independent of path.

Let $f(z)$ be analytic in a region 'R' and on its boundary 'C'.
Then $\oint_C f(z) dz = 0$. This fundamental theorem often
called Cauchy-Goursat Theorem.

**See Example on
Page 27, 28, 29**

Integrals (Recall)

$$w(t) = u(t) + i v(t)$$

Reading

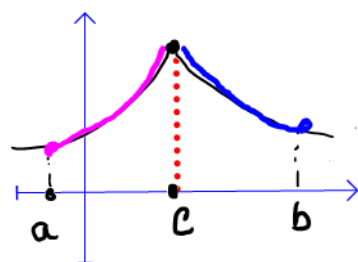
t is a real variable
 $w(t)$ is a complex valued function

$$\Rightarrow w'(t) = u'(t) + i v'(t)$$

$$\Rightarrow \frac{d}{dt} [z_0 w(t)] = z_0 w'(t) \quad z_0 = x_0 + i y_0$$

$$\Rightarrow \frac{d}{dt} [e^{z_0 t}] = z_0 e^{z_0 t}$$

$$\Rightarrow \int_a^b w(t) dt = \int_a^b u(t) dt + i \int_a^b v(t) dt$$



for $c \in [a, b]$

$$\begin{aligned} \Rightarrow \int_a^b w(t) dt &= \int_a^c w(t) dt + \int_c^b w(t) dt \\ &= \int_a^c [u(t) + i v(t)] dt + \int_c^b [u(t) + i v(t)] dt \end{aligned}$$

Examples:

1. Evaluate $\int_1^2 \left(\frac{1}{t} - i\right)^2 dt$ $\leftarrow$ Definite Integral

$$\begin{aligned} &\int_1^2 \left(\frac{1}{t} - i\right)^2 dt \\ &= \int_1^2 \left[\frac{1}{t^2} - 2i \cdot \frac{1}{t} + i^2 \right] dt \\ &= \int_1^2 \left(\frac{1}{t^2} - 1 \right) dt - 2i \int_1^2 \frac{1}{t} dt \\ &= \left[-\frac{1}{t} - t \right]_1^2 - i \left[2 \ln t \right]_1^2 \\ &= \left[\left(-\frac{1}{2} - 2\right) - (-1 - 1) \right] - i \left[2 \ln 2 - 2 \ln 1 \right] \\ &= -\frac{1}{2} - 2 + 2 - i \left(2 \ln \left(\frac{2}{1} \right) \right) = -\frac{1}{2} - i 2 \ln 2 \end{aligned}$$

$$\begin{aligned} &\int t^{-2} dt \\ &= t^{-2+1} \\ &= \frac{t^{-1}}{-1} \end{aligned}$$

$$2. \int_0^{\infty} e^{-zt} dt \quad (\operatorname{Re} z > 0) \leftarrow \text{Improper Integral}$$

$$= \lim_{l \rightarrow \infty} \int_0^l e^{-zt} dt \quad l \in \mathbb{C}$$

$$= \lim_{l \rightarrow \infty} \left[\frac{e^{-zt}}{-z} \right]_0^l$$

$$= \lim_{l \rightarrow \infty} \left[-\frac{e^{-zl}}{z} + \frac{e^0}{z} \right] \begin{cases} -zl \\ = -z\infty \\ = -\infty \end{cases} \text{ as } l \rightarrow \infty$$

$$= -\frac{e^{-\infty}}{z} + \frac{1}{z}$$

$$= -\frac{1}{ze^{\infty}} + \frac{1}{z}$$

$$= -\frac{1}{\infty} + \frac{1}{z}$$

$$= 0 + \frac{1}{z}$$

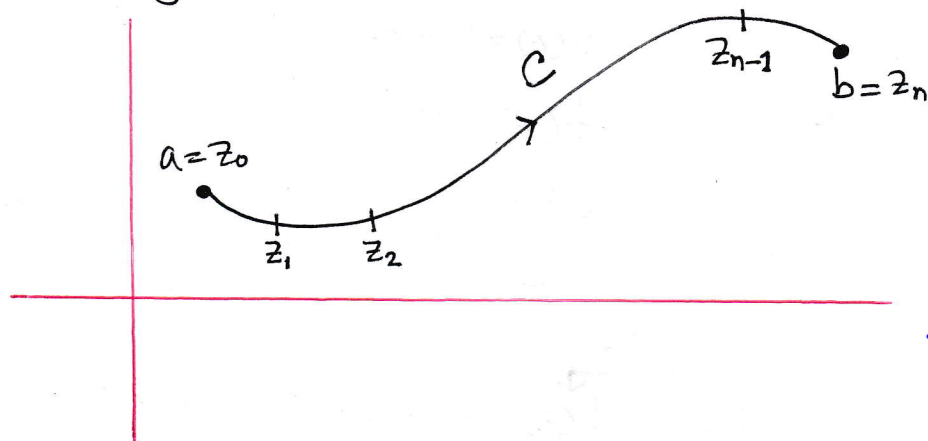
$$= \frac{1}{z}$$

Complex Line Integral

The definite integral of $f(z)$ from a to b along curve C is:

$$\int_a^b f(z) dz \quad \text{or} \quad \int_C f(z) dz$$

It is also known as path integral, contour integral, curve integral.



Parametric eqn:

$$z = z(t)$$

$$\therefore dz = z'(t) dt$$

$$z = z(t)$$

$$\therefore f(z) = f(z(t))$$

$$\int_a^b f(z) dz = \int_C \underbrace{f(z)}_{f(z(t))} \underbrace{dz}_{z'(t) dt} = \int_{t_1}^{t_2} f(z(t)) z'(t) dt$$

$$= \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2} dt$$

See page 15-16, 20-27, 30

(considering parametric eqn
 $z = z(t) = x(t) + iy(t)$)

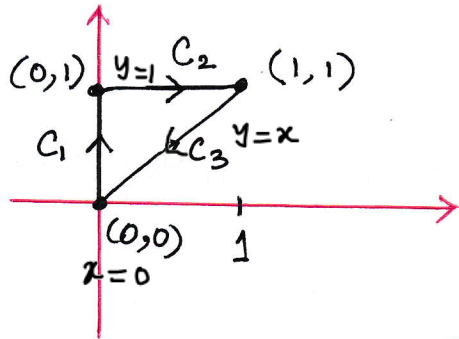
$$t_1 \leq t \leq t_2$$

Theorem $\longrightarrow$ PAGE 8

$f(z)$ is continuous on C . let $z(t)$ be any parametric function of C for $a \leq t \leq b$, then
 $\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt$ $\longrightarrow$ (It is known as chain rule)
 $= \int_a^b |z'(t)| dt = \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2} dt = \text{arc length}$

Examples:

① Evaluate $\int_C f(z) dz$ over the close triangular contour with vertices $(0,0)$, $(0,1)$, $(1,1)$ where $f(z) = y - x - 3ix^2$

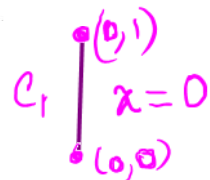


$C_1 + C_2 \gg C_3$ (triangular inequality)

$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz + \left(- \int_{C_3} f(z) dz \right)$$

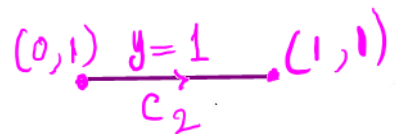
$C_1: x=0 \Rightarrow z = iy$ $f(z) = y - x - 3ix^2 = y - 0 - 3i(0)^2 = y$
 $dz = i dy$ $f(z) = y$

$$\int_{C_1} f(z) dz = \int_0^1 y i dy = i \left[\frac{y^2}{2} \right]_0^1 = \frac{i}{2}$$



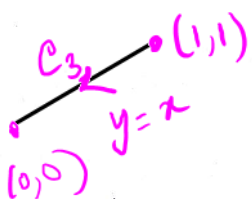
$C_2: y=1 \Rightarrow z = x + iy = x + i$ $f(z) = y - x - 3ix^2 = 1 - x - 3ix^2$
 $dz = dx$

$$\int_{C_2} f(z) dz = \int_0^1 (1 - x - 3ix^2) dx = \left[x - \frac{x^2}{2} - 3i \frac{x^3}{3} \right]_0^1 = 1 - \frac{1}{2} - i = \frac{1}{2} - i$$



$C_3: y=x \Rightarrow z = x + iy = x + ix$ $f(z) = y - x - 3ix^2 = x - x - 3ix^2 = -3ix^2$
 $z = (1+i)x$ $f(z) = -3ix^2$
 $dz = (1+i)dx$

$$\int_{C_3} f(z) dz = \int_0^1 (-3ix^2)(1+i) dx = \int_0^1 (-3ix^2 - 3i^2 x^2) dx = \int_0^1 (-3ix^2 + 3x^2) dx$$



$$= \int_0^1 (-3ix^2 + 3x^2) dx$$

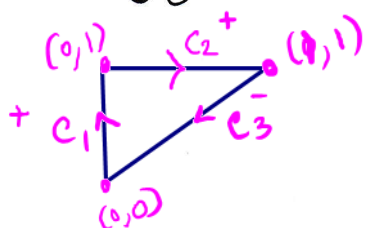
$$= \int_0^1 3x^2(1-i) dx$$

$$= 3(1-i) \left[\frac{x^3}{3} \right]_0^1$$

$$= (1-i)(1-0)$$

$$= 1-i$$

$$\therefore \int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz + \left[\int_{C_3} f(z) dz \right]$$



$$= \left[\frac{i}{2} \right] + \left[\frac{1}{2} - i \right] + \left[-(1-i) \right]$$

$$= \frac{i}{2} - \frac{1}{2} = \frac{1}{2}(-1+i)$$

② Evaluate $\int_C \bar{z} dz$

(a) from $z=0$ to $z=4+2i$ along the curve c given by

$$z = t^2 + it$$

(b) the line from $z=0$ to $z=2i$ and then the line from $z=2i$ to $z=4+2i$

a) when $z=0$; $\rightarrow 0 = 0+0i = t^2 + it$
by equating factor of like terms: $t=0$

when $z=4+2i$; $4+2i = t^2 + it$
by equating factor of like terms: $t=2$

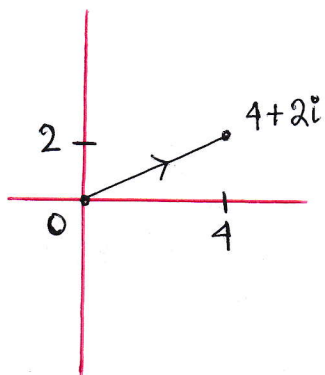
$$\therefore 0 \leq t \leq 2 \quad \text{or} \quad t \in [0, 2]$$

$$z = t^2 + it \Rightarrow dz = (2t + i) dt \xrightarrow{\text{red}} z'(t)$$

$$\bar{z} = t^2 - it = f(z(t)) \quad \because f(z) = \bar{z} \text{ (given)}$$

$$\int_C f(z) dz = \int_C \bar{z} dz = \int_C f(z(t)) z'(t) dt$$

$$= \int_0^2 (t^2 - it)(2t + i) dt = \int_0^2 (2t^3 + it^2 - 2it^2 - i^2 t) dt$$

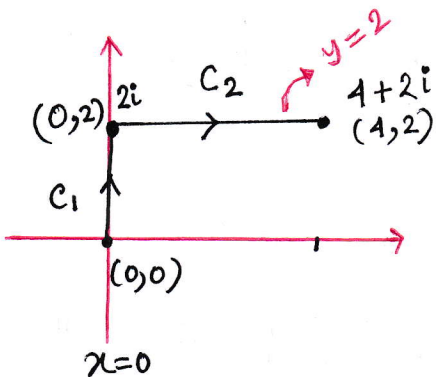


$$= \int_0^2 (2t^3 - it^2 + t) dt$$

$$= \left[\frac{2t^4}{4} - \frac{it^3}{3} + \frac{t^2}{2} \right]_0^2$$

$$= \frac{16}{2} - i \frac{8}{3} + \frac{4}{2} - 0 = 10 - \frac{8}{3}i$$

b)



$$\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz$$

$$C_1: x=0, \Rightarrow z = x + iy \Rightarrow \bar{z} = -iy = f(z)$$

$$dz = i dy$$

$$0 \leq y \leq 2$$

$$\int_{C_1} f(z) dz = \int_{C_1} \bar{z} dz = \int_0^2 (-iy) i dy = \int_0^2 -i^2 y dy = \left[\frac{y^2}{2} \right]_0^2 = 2$$

$$C_2: y=2, \quad z = x + iy \Rightarrow \bar{z} = x - 2i = f(z)$$

$$z = x + 2i \Rightarrow \bar{z} = x - 2i$$

$$0 \leq x \leq 4 \quad dz = dx$$

$$\int_{C_2} f(z) dz = \int_{C_2} \bar{z} dz = \int_0^4 (x - 2i) dx$$

$$= \left[\frac{x^2}{2} - 2ix \right]_0^4$$

$$= 8 - 8i$$

$$\int_C f(z) dz = \int_C \bar{z} dz = \int_{C_1} dz + \int_{C_2} dz = 2 + 8 - 8i = 10 - 8i$$

③ Evaluate $\int_c \bar{z} dz$ where c is a right handed half of circle $|z|=2$ or $z=2e^{i\theta}$, $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.

The contour is given by

$$|z|=2$$

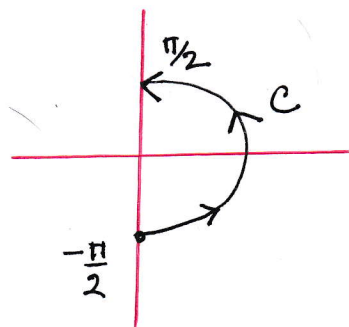
$$\text{or } |z|=R$$

$$z=2e^{i\theta}$$

$$\Rightarrow z=re^{i\theta}$$

$$\bar{z}=2e^{-i\theta}$$

$$\int_c f(z) dz$$



$$\begin{cases} z=2e^{i\theta} \\ \Rightarrow dz=2ie^{i\theta}d\theta \\ \rightarrow dz=2e^{i\theta}i d\theta \end{cases}$$

$$= \int_c \bar{z} dz$$

$$= \int_{-\pi/2}^{\pi/2} (2e^{-i\theta})(2ie^{i\theta}d\theta)$$

$$= 4i \left[\theta \right]_{-\pi/2}^{\pi/2} = 4i \left[\frac{\pi}{2} - \left(-\frac{\pi}{2}\right) \right] = 4\pi i$$

(Exercise Sheet #4)

① A counter clockwise oriented contour is called a positively oriented contour.

Evaluate $\int_{(0,1)}^{(2,5)} (3x+y)dx + (2y-x)dy$ along

- (a) the curve $y=x^2+1$
- (b) the straight line joining $(0,1)$ & $(2,5)$
- (c) the straight lines from $(0,1)$ to $(0,5)$ and then from $(0,5)$ to $(2,5)$.
- (d) the straight lines from $(0,1)$ to $(2,1)$ and then from $(2,1)$ to $(2,5)$.

(a) $y = x^2 + 1 \rightarrow \text{Given}$

$$dy = 2x dx$$

$$\int_{(0,1)}^{(2,5)} ((3x+y) dx + (2y-x) dy)$$

$$= \int_0^2 ((3x + x^2 + 1) dx + (2(x^2 + 1) - x) 2x dx)$$

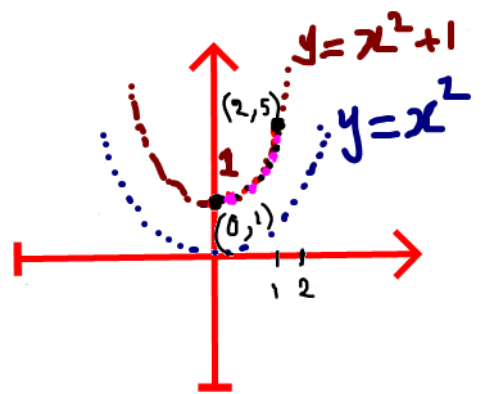
$$= \int_0^2 ((3x + x^2 + 1) dx + (2x^2 + 2 - x) 2x dx)$$

$$= \int_0^2 (3x + x^2 + 1 + 4x^3 + 4x - 2x^2) dx$$

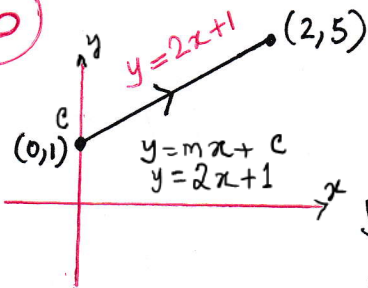
$$= \int_0^2 (4x^3 - x^2 + 7x + 1) dx$$

$$= \left[\frac{4x^4}{4} - \frac{x^3}{3} + \frac{7x^2}{2} + x \right]_0^2$$

$$= 16 - \frac{8}{3} + 7 \cdot \frac{4}{2} + 2 = 16 - \frac{8}{3} + 14 + 2 = \frac{88}{3}$$



(b)



$$(x_1, y_1) = \begin{cases} (0,1) \\ \text{or } (2,5) \end{cases}$$

$$m = \frac{dy}{dx} = \frac{4}{2} = 2$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 2(x - 0)$$

$$y = 2x + 1$$

$$dy = 2 dx$$

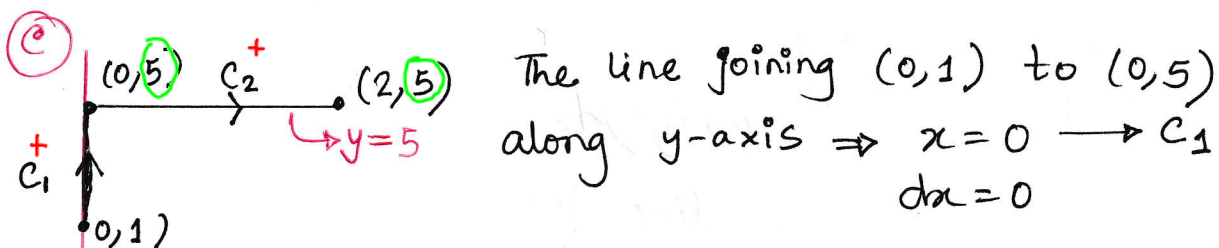
$$\int_{(0,1)}^{(2,5)} ((3x+y) dx + (2y-x) dy)$$

$$= \int_0^2 ((3x + 2x + 1) dx + (2(2x + 1) - x) 2 dx)$$

$$= \int_0^2 ((5x + 1) + (6x + 4)) dx$$

$$= \int_0^2 (11x + 5) dx = \left[11 \cdot \frac{x^2}{2} + 5x \right]_0^2$$

$$= 11 \cdot \frac{4}{2} + 5(2) - 0 = 32$$



$$\int_C = \int_{C_1} + \int_{C_2}$$

$$C_1 = \int_{(0,1)}^{(2,5)} (3x+y)dx + (2y-x)dy$$

$$= \int_1^5 (0+y) \cdot 0 + (2y-0)dy$$

$$= \int_1^5 2y dy = 2 \left[\frac{y^2}{2} \right]_1^5 = 25 - 1 = 24$$

The line joining (0,5) to (2,5) is:

$$x\text{-axis: } y=5 \longrightarrow C_2$$

$$dy=0$$

$$C_2: \int_{(0,1)}^{(2,5)} (3x+y)dx + (2y-x)dy$$

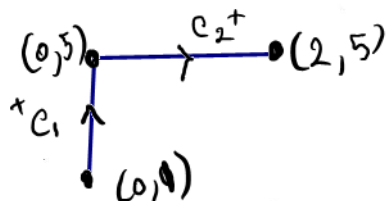
$$= \int_0^2 (3x+5)dx + (10-x)(0)$$

$$= \int_0^2 (3x+5)dx$$

$$= \left[\frac{3x^2}{2} + 5x \right]_0^2 = \frac{3}{2} \cdot 4 + 5(2) - 0 = 16$$

$$\therefore \text{The required value} = \oint_C f(z)dz = \int_{C_1} \dots dz + \int_{C_2} \dots dz$$

$$= 24 + 16 = 40$$



(Exercise Sheet #4)

6. Evaluate $\int_i^{2-i} (3xy + iy^2) dz$

a) along the straight line joining $z=i$ and $z=2-i$

b) along the parabola $x=2t-2, y=1+t-t^2$

$$\begin{aligned} \text{a) } z=i &\Rightarrow (0,1) & m &= \frac{dy}{dx} \\ z=2-i &\Rightarrow (2,-1) & &= -\frac{2}{2} = -1 \end{aligned}$$

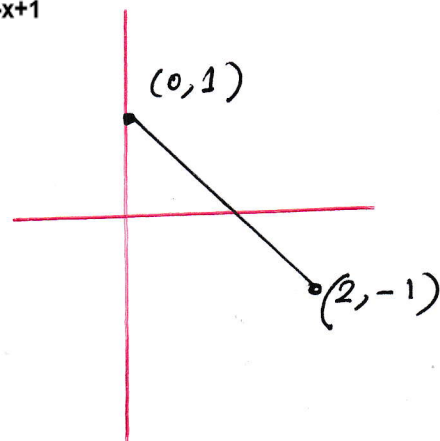
$$z = x + iy \quad y - y_1 = m(x - x_1)$$

$$dz = dx + i dy \quad y - 1 = -1(x - 0)$$

$$\begin{aligned} y - 1 &= -x \\ y - 1 &= -x \\ dy &= -dx \end{aligned}$$

$$y = mx + c \quad m = -1, c = 1$$

$$\text{so, } y = -x + 1$$



$$\int_i^{2-i} (3xy + iy^2) dz$$

$$\int_{(0,1)}^{(2,-1)} (3xy + iy^2) (dx + i dy)$$

$$= \int_0^2 \{ 3x(-x+1) + i(-x+1)^2 \} \{ dx + i(-dx) \}$$

$$= \int_0^2 \{ -3x^2 + 3x + i(x^2 - 2x + 1) \} (1-i) dx$$

$$= \int_0^2 (-3x^2 + 3x + ix^2 - 2ix + i) (1-i) dx$$

$$= (1-i) \left[-\frac{3x^3}{3} + \frac{3x^2}{2} + i\frac{x^3}{3} - 2i\frac{x^2}{2} + ix \right]_0^2$$

$$= (1-i) [-8 + 6 + i \frac{8}{3} - 4i + 2i - 0]$$

$$= (1-i) (-8 + 6 + i (\frac{8-12+6}{3}))$$

$$= (1-i) (-2 + \frac{2i}{3}) = -2 + 2i + \frac{2}{3}i - \frac{2}{3}i^2 = \frac{8i}{3} - \frac{4}{3}$$

$$= \frac{4}{3} (-1 + 2i)$$

$$6b) \int_i^{2-i} (3xy + iy^2) dz$$

$$x = 2t - 2$$

$$y = 1 + t - t^2$$

$$= \int_{(0,1)}^{(2,-1)} (3xy + iy^2) dz$$

$$0 \leq x \leq 2 \quad \& \quad 1 \leq y \leq -1$$

$$0 \leq 2t - 2 \leq 2 \quad 1 \leq (1 + t - t^2) \leq -1$$

$$2t - 2 = 0 \quad ; \quad 2t - 2 = 2$$

$$\Rightarrow 2t = 2 \quad \Rightarrow 2t = 2 + 2 = 4$$

$$\Rightarrow t = 1 \quad \Rightarrow t = 2$$

$$\int_1^2 f(z(t)) z'(t) dt$$

$$z = z(t)$$

$$f(z) = f(z(t))$$

$$\frac{dz}{dt} = z'(t) dt$$

$$\int_{(0,1)}^{(2,-1)} = \int_{t=1}^2$$

$$= \int_1^2 \left[-6t^3 + 12t^2 - 6 + i(t^4 - 2t^3 - t^2 + 2t + 1) \right] [2 + i(1 - 2t)] dt$$

$$f(z) = 3xy + iy^2$$

$$f(z(t)) = 3(2t - 2)(1 + t - t^2) + i(1 + t - t^2)^2$$

$$= \int_1^2 (-6t^3 + 12t^2 - 6 + i t^4 - i 2t^3 - t^2 + i 2t + i) (2 + i - 2it) dt$$

$$= (6t - 6)(1 + t - t^2) + i((1 + t) - t^2)^2$$

$$= (6t - 6 + 6t^2 - 6t - 6t^3 + 6t^2) + i[(1 + t)^2 - 2(1 + t)t^2 + t^4]$$

$$= \int_1^2 \left(\boxed{-12t^3} + \boxed{24t^2} - \boxed{12} + \boxed{2it^4} - \boxed{4it^3} + \boxed{4it} + \boxed{2i} \right) dt$$

$$= (-6t^3 + 12t^2 - 6) + i[1 + 2t + t^2 - 2t^2 - 2t^3 + t^4]$$

$$= (-6t^3 + 12t^2 - 6) + i(t^4 - 2t^3 - t^2 + 2t + 1)$$

$$= \int_1^2 \left[-5t^4 - 12t^3 + 29t^2 - 13 \right] + i \left[14t^4 - 34t^3 + 10t^2 + 16t - 4 \right] dt$$

$$z = x + iy$$

$$z(t) = (2t - 2) + i(1 + t - t^2)$$

$$z'(t) = 2 + i(1 - 2t)$$

$$= \left[-\frac{5t^5}{5} - \frac{12t^4}{4} + \frac{29t^3}{3} - 13t \right]_1^2 + i \left[\frac{14t^5}{5} - \frac{34t^4}{4} + \frac{10t^3}{3} + \frac{16t^2}{2} - 4t \right]_1^2$$

$$= \left[(-32 - 48 + \frac{232}{3} - 26) - (-1 - 3 + \frac{29}{3} - 13) \right] + i \left[\left(\frac{448}{5} - 136 + \frac{80}{3} + 32 - 8 \right) - \left(\frac{14}{5} - \frac{17}{2} + \frac{10}{3} + 8 - 4 \right) \right]$$

$$= -21.33 + i2.63$$

(Exercise sheet #4)

8. Evaluate $\int_C \frac{dz}{z-2}$ around the circle $|z-2|=4$

$$|z-2|=4$$

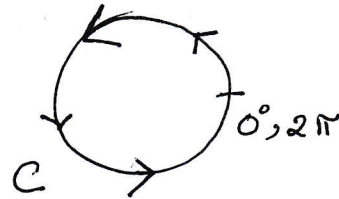
$$z-2 = 4e^{i\theta}$$

$$z = 2 + 4e^{i\theta}$$

$$dz = 4ie^{i\theta} d\theta$$

$$|z|=R$$

$$z = Re^{i\theta}$$

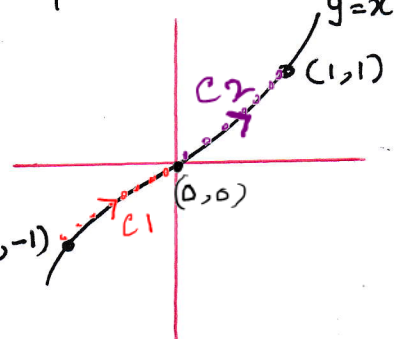


$$\int \frac{dz}{z-2} = \int_0^{2\pi} \frac{4ie^{i\theta}}{4e^{i\theta}} d\theta = i \int_0^{2\pi} e^0 d\theta = i \int_0^{2\pi} 1 d\theta = i [\theta]_0^{2\pi} = 2\pi i$$

Example $f(z)$ is defined by equations

$$f(z) = \begin{cases} 1, & y < 0 \\ 4y, & y > 0 \end{cases} \quad \begin{array}{l} c \text{ is the arc from} \\ z = -1-i \text{ to } z = 1+i \\ \text{along the curve } y = x^3 \end{array}$$

$f(z)$ is changing the ^{concavity} at $y=0$ $\therefore$ the conditions are provided as $y > 0$ or $y < 0$
 $z = -1, -i \Rightarrow (-1, -1); z = 1+i \Rightarrow (1, 1)$ which implies $y \neq 0$

$$\therefore \int_c f(z) dz = \int_{c_1}^{(0,0)} f(z) dz + \int_{(0,0)}^{(1,1)} f(z) dz$$


$c_1: y = x^3, z = x + iy = x + ix^3 \quad \therefore y = x^3$
 $\downarrow$
 $(-1, -1) \text{ to } (0, 0) \quad dz = (1 + 3x^2i) dx \quad \text{over } -1 \leq x \leq 0.$
 when $y < 0$

$$\begin{aligned} \int_{c_1} f(z) dz &= \int_{c_1} 1 dz = \int_{-1}^0 (1 + 3x^2i) dx \\ &= \left[x + \frac{3x^3i}{3} \right]_{-1}^0 \\ &= -(-1 - i) = 1 + i \end{aligned}$$

$C_2: (0,0) \text{ to } (1,1)$

$$\int_{C_2} f(z) dz = \int_{C_2} 4y dz$$

$$= \int_0^1 4x^3 (1+3x^2i) dx$$

$$y = x^3$$

$$z = x + iy$$

$$= x + ix^3$$

$$dz = (1+3x^2i) dx$$

$$= \int_0^1 (4x^3 + 12x^5i) dx$$

$$= \left[\cancel{4} \frac{x^4}{\cancel{4}} + i \cancel{12} \frac{x^6}{\cancel{6}} \right]_0^1$$

$$= \left[x^4 + 2ix^6 \right]_0^1$$

$$= 1 + 2i$$

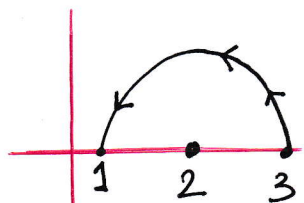
$$\therefore \int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz$$

$$= 1+i + 1+2i$$

$$= 2+3i$$

Example Evaluate the contour integral $\int_c \frac{dz}{z-2}$

where c is the upper semicircle with radius 1 centered at $z=2$. $\rightarrow 0^\circ$ to π



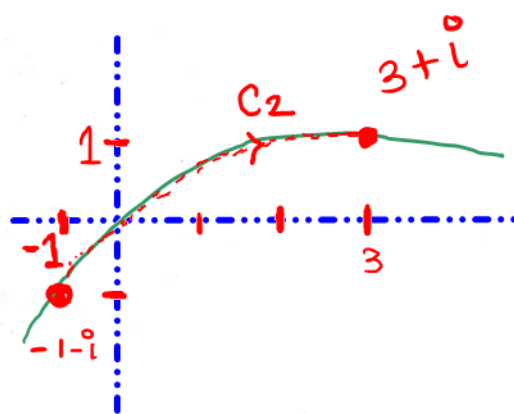
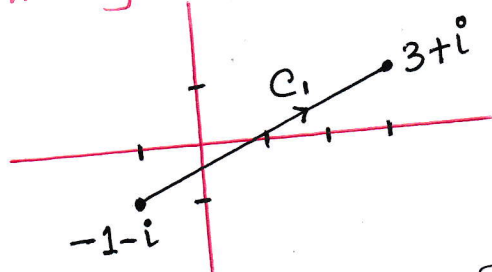
$$\int_c \frac{dz}{z-2} = \int_0^\pi \frac{e^{i\theta} \cdot i d\theta}{e^{i\theta}}$$

$$\begin{aligned} |z-z_0| &= r \\ z-z_0 &= r e^{i\theta} \\ z &= z_0 + r e^{i\theta} \\ &= 2 + 1 e^{i\theta} \\ z &= 2 + e^{i\theta}, \quad 0 \leq \theta \leq \pi \\ dz &= i e^{i\theta} d\theta \end{aligned}$$

$$\begin{aligned} |z-2| &= r \\ z-2 &= r e^{i\theta} \\ z-2 &= 1 e^{i\theta} \\ z &= 2 + e^{i\theta} \\ dz &= e^{i\theta} i d\theta \end{aligned}$$

$$\begin{aligned} \int_c \frac{1}{z-2} dz &= \int_0^\pi \frac{i e^{i\theta}}{2 + e^{i\theta} - 2} d\theta \\ &= \int_0^\pi i d\theta \\ &= i [\theta]_0^\pi = \pi i \end{aligned}$$

Example Show that $\int_{c_1} z dz = \int_{c_2} z dz = 4 + 2i$,
where c_1 is the line segment from $-1-i$ to $3+i$,
and c_2 is the portion of the parabola $x = y^2 + 2y$
joining $-1-i$ to $3+i$



a) The line segment that contains the points $(-1, -1), (3, 1)$

$$m = \frac{\Delta y}{\Delta x} = \frac{2}{4} = \frac{1}{2}$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{1}{2}(x - 3)$$

$$y = 1 + \frac{x}{2} - \frac{3}{2}$$

$$y = \frac{1}{2}x - \frac{1}{2}$$

OR $x = 2y + 1$

along C_1 : $z = x + iy = 2y + 1 + iy$
 $dz = (2 + i)dy$

$$\int_{C_1} z dz = \int_{-1}^1 (2y + 1 + iy)(2 + i) dy$$

$$= \int_{-1}^1 (4y + 2 + 2iy + 2iy + i - y) dy$$

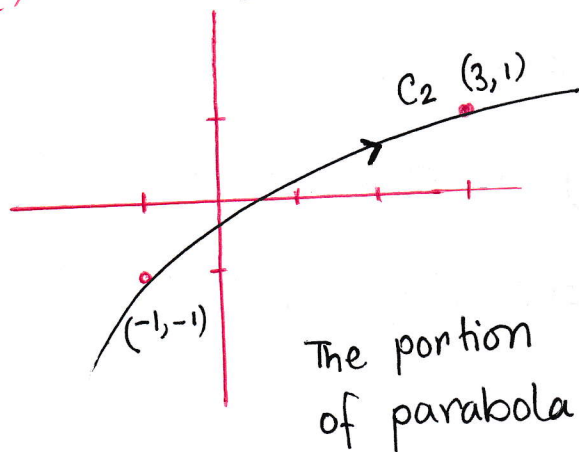
$$= \int_{-1}^1 (3y + 4iy + i + 2) dy$$

$$= \left[\frac{3y^2}{2} + 4i \frac{y^2}{2} + iy + 2y \right]_{-1}^1$$

$$= \frac{3}{2} + 2i + i + 2 - \left(\frac{3}{2} + 2i - i - 2 \right)$$

$$= 4 + 2i$$

(b) similarly along C_2



$$z = x + iy$$

$$= (y^2 + 2y) + iy \quad \text{given } x = y^2 + 2y$$

$$dz = (2y + 2 + i) dy$$

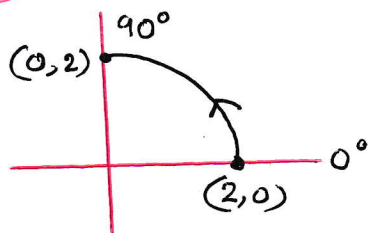
$$\begin{aligned} \int_{C_2} z dz &= \int_{-1}^1 [(y^2 + 2y) + iy] (2y + 2 + i) dy \\ &= \int_{-1}^1 (2y^3 + 4y^2 + 2iy^2 + 2y^2 + 4y + 2iy + iy^2 + 2iy - y) dy \\ &= \int_{-1}^1 (2y^3 + 6y^2 + 3iy^2 + 3y + 4iy) dy \\ &= \left[\frac{2y^4}{4} + \frac{6y^3}{3} + 3i \frac{y^3}{3} + \frac{3y^2}{2} + 4i \frac{y^2}{2} \right]_{-1}^1 \\ &= \frac{1}{2} + 2 + i + \frac{3}{2} + 2i - \left(\frac{1}{2} - 2 - i + \frac{3}{2} + 2i \right) \\ &= 4 + 2i \end{aligned}$$

This outcome does not hold in general for arc length while C_1, C_2 are independent of paths.

(Exercise Sheet #4)

5. Evaluate $\oint_C (z^2 + 3z) dz$

a) along the circle $|z|=2$ from $(2,0)$ to $(0,2)$ in a counter clockwise direction.



$$|z|=2 \Rightarrow z = 2e^{i\theta} \\ \therefore dz = 2ie^{i\theta} d\theta$$

$$\begin{aligned} \oint_C (z^2 + 3z) dz &= \int_0^{\pi/2} (4e^{2i\theta} + 6e^{i\theta}) 2ie^{i\theta} d\theta \\ &= 2i \int_0^{\pi/2} (4e^{3i\theta} + 6e^{2i\theta}) d\theta \\ &= 2i \left[\frac{4e^{3i\theta}}{3i} + \frac{6e^{2i\theta}}{2i} \right]_0^{\pi/2} \\ &= 2 \left[\frac{4}{3} e^{3i\theta} + 3e^{2i\theta} \right]_0^{\pi/2} \\ &= 2 \left[\left(\frac{4}{3} e^{i\frac{3\pi}{2}} + 3e^{i\pi} \right) - \left(\frac{4}{3} e^0 + 3e^0 \right) \right] \\ &= 2 \left[\left\{ \frac{4}{3} \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) + 3(\cos \pi + i \sin \pi) \right\} \right. \\ &\quad \left. - \left\{ \frac{4}{3} + 3 \right\} \right] \\ &= 2 \left[\frac{4}{3} (0 - i) + 3(-1 - 0) - \frac{4}{3} - 3 \right] \\ &= 2 \left[-\frac{4}{3}i - 3 - \frac{13}{3} \right] \\ &= 2 \left[-\frac{4}{3}i - \frac{22}{3} \right] = -\frac{44}{3} - \frac{8}{3}i \end{aligned}$$